Binance Gold & Silver Perpetuals and ETF Markets

KSE Python course project

1. Introduction and research question

Research question.
Did the introduction of Binance TRADFI gold and silver perpetual futures change volatility, liquidity and price integration in the corresponding precious metal ETF markets (GLD and SLV)?

Main hypotheses (testable with pre/post data).

- **H1 (volatility)**: Realized volatility of gold and silver ETFs increased after the launch of Binance XAUUSDT and XAGUSDT TRADFI perpetual futures.
- **H2 (liquidity)**: Trading activity and liquidity of GLD and SLV improved (higher volume, lower Amihud illiquidity) after the launch.

We do *not* formulate a formal hypothesis on "integration increased", because Binance returns do not exist before the launch, so we cannot compute pre-event GLD–XAU or SLV–XAG correlation or beta. We still report post-event rolling correlation and integration metrics as **descriptive** evidence (how strongly ETF and perpetual co-move after the launch), but we do not test a pre/post change for integration.

2. Data and feature set

We use three pre-built datasets produced by `build_features.py`:

- **Daily panel** (`features_daily.csv`): 112 trading days, 85 variables. Used for the event study (pre/post comparison, regressions).
- **Hourly panel** (`features_hourly.csv`): one row per hour over the same calendar span; Binance 1m aggregated to hourly, ETF/controls forward-filled from daily.
- **1-minute panel** (`features_1m.csv`): one row per Binance XAUUSDT 1m candle with daily ETF/controls merged by date. Used to satisfy the course requirement of **10,000 observations**.

Raw data sources and variable definitions are in `DATA_DESCRIPTION.md`, `VARIABLES_DESCRIPTION.md`, and `DATA_ROW_COUNTS_AND_PERIODS.md`.

```
Daily: 118 rows (features_daily.csv)
Hourly: 4104 rows (features_hourly.csv)
1-minute: 67074 rows (features_1m.csv)
```

```
(118, 87)
```

Date	px_gld_open	px_gld_high	px_gld_low	px_gld_close	vol_gld	px_slv_open	px_
2025-09-02	320.820007	325.920013	320.239990	325.589996	21247500.0	36.630001	37.
2025-09-03	327.760010	329.450012	326.730011	328.140015	16062200.0	37.330002	37.
2025-09-04	327.029999	327.559998	325.350006	326.690002	11703100.0	37.200001	37.
2025-09-05	329.529999	331.440002	328.929993	331.049988	16065200.0	37.410000	37.
2025-09-08	333.660004	335.670013	333.230011	334.820007	18617300.0	37.570000	37.

2.1 Basic cleaning and event date

We define the **event** as the **launch** of Binance XAUUSDT TRADFI perpetual futures. So there is one date: 2026-01-05 (event = launch).
To keep a meaningful pre-event window, we only require that ETF prices and returns are available; Binance-specific fields are allowed to be missing before the launch.

```
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 117 entries, 2025-09-03 to 2026-02-19
Data columns (total 87 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   px_gld_open                     117 non-null    float64
 1   px_gld_high                     117 non-null    float64
 2   px_gld_low                      117 non-null    float64
 3   px_gld_close                    117 non-null    float64
 4   vol_gld                         117 non-null    float64
 5   px_slv_open                     117 non-null    float64
 6   px_slv_high                     117 non-null    float64
 7   px_slv_low                      117 non-null    float64
 8   px_slv_close                    117 non-null    float64
 9   vol_slv                         117 non-null    float64
 10  dxy                             117 non-null    float64
 11  vix                             117 non-null    float64
 12  tnx                             117 non-null    float64
 13  dgs10                           114 non-null    float64
 14  px_xau_binance_open             32 non-null     float64
 15  px_xau_binance_high             32 non-null     float64
 16  px_xau_binance_low              32 non-null     float64
 17  px_xau_binance_close            32 non-null     float64
 18  px_xau_binance_volume           32 non-null     float64
```

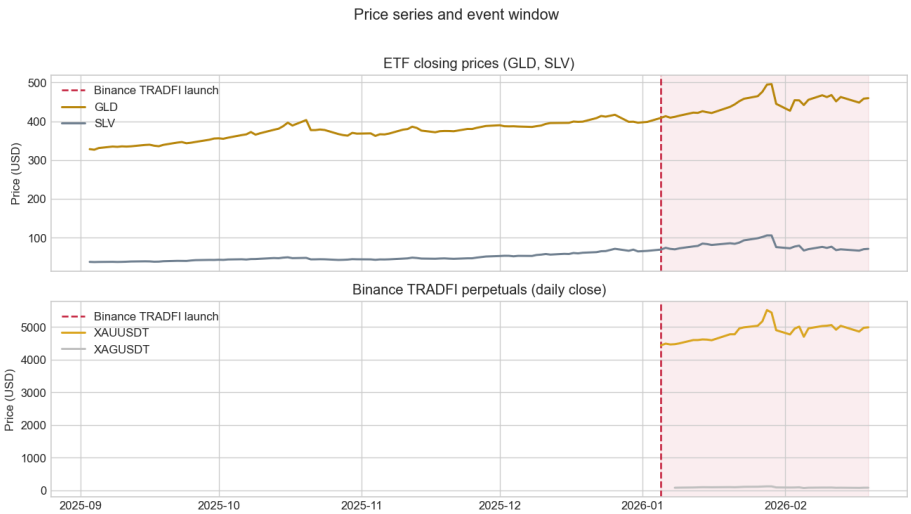
19	px_xau_binance_num_trades	32 non-null	float64
20	px_xag_binance_open	29 non-null	float64
21	px_xag_binance_high	29 non-null	float64
22	px_xag_binance_low	29 non-null	float64
23	px_xag_binance_close	29 non-null	float64
24	px_xag_binance_volume	29 non-null	float64
25	px_xag_binance_num_trades	29 non-null	float64
26	funding_xau	22 non-null	float64
27	funding_mark_xau	22 non-null	float64
28	funding_xag	22 non-null	float64
29	funding_mark_xag	22 non-null	float64
30	oi_xau	0 non-null	float64
31	oi_xag	0 non-null	float64
32	ret_xau_binance	31 non-null	float64
33	ret_xag_binance	28 non-null	float64
34	ret_gld	117 non-null	float64
35	ret_slv	117 non-null	float64
36	ret_xau_binance_abs	31 non-null	float64
37	ret_xau_binance_sq	31 non-null	float64
38	ret_xag_binance_abs	28 non-null	float64
39	ret_xag_binance_sq	28 non-null	float64
40	ret_gld_abs	117 non-null	float64
41	ret_gld_sq	117 non-null	float64
42	ret_slv_abs	117 non-null	float64
43	ret_slv_sq	117 non-null	float64
44	vol_realized_10_ret_xau_binance	22 non-null	float64
45	vol_realized_20_ret_xau_binance	12 non-null	float64
46	vol_realized_10_ret_xag_binance	19 non-null	float64
47	vol_realized_20_ret_xag_binance	9 non-null	float64
48	vol_realized_10_ret_gld	108 non-null	float64
49	vol_realized_20_ret_gld	98 non-null	float64
50	vol_realized_10_ret_slv	108 non-null	float64
51	vol_realized_20_ret_slv	98 non-null	float64
52	vol_parkinson_gld	99 non-null	float64
53	vol_parkinson_slv	99 non-null	float64
54	vol_parkinson_xau_binance	13 non-null	float64
55	vol_parkinson_xag_binance	10 non-null	float64
56	vol_gk_gld	99 non-null	float64
57	vol_gk_slv	99 non-null	float64
58	vol_gk_xau_binance	13 non-null	float64
59	vol_gk_xag_binance	10 non-null	float64
60	corr_20_gld_xau	12 non-null	float64
61	corr_20_slv_xag	9 non-null	float64
62	beta_60_gld_on_xau	0 non-null	float64
63	beta_60_slv_on_xag	0 non-null	float64
64	vol_ratio_gld_xau	12 non-null	float64

```
 65  vol_ratio_slv_xag              9 non-null      float64
 66  abs_return_gap_gld_xau         31 non-null     float64
 67  abs_return_gap_slv_xag         28 non-null     float64
 68  post_event                     117 non-null    int64
 69  event_window_7                 117 non-null    int64
 70  event_window_14                117 non-null    int64
 71  log_vol_gld                    117 non-null    float64
 72  log_vol_slv                    117 non-null    float64
 73  amihud_gld                     117 non-null    float64
 74  amihud_20_gld                  98 non-null     float64
 75  amihud_slv                     117 non-null    float64
 76  amihud_20_slv                  98 non-null     float64
 77  funding_xau_change             21 non-null     float64
 78  funding_extreme_xau            117 non-null    int64
 79  oi_xau_change                  0 non-null      float64
 80  funding_xag_change             21 non-null     float64
 81  funding_extreme_xag            117 non-null    int64
 82  oi_xag_change                  0 non-null      float64
 83  dxy_ret                        117 non-null    float64
 84  vix_ret                        117 non-null    float64
 85  tnx_ret                        117 non-null    float64
 86  dgs10_ret                      112 non-null    float64
dtypes: float64(82), int64(5)
memory usage: 80.4 KB
```

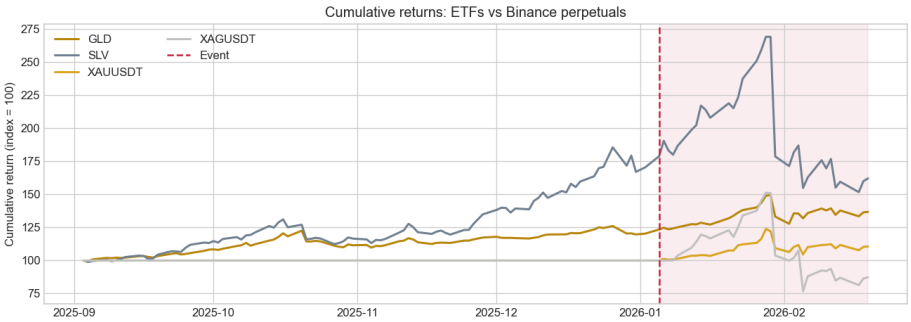
```
post_event
0             85
1             32
Name: count, dtype: int64
```

3. Descriptive statistics

3.1 Prices and returns



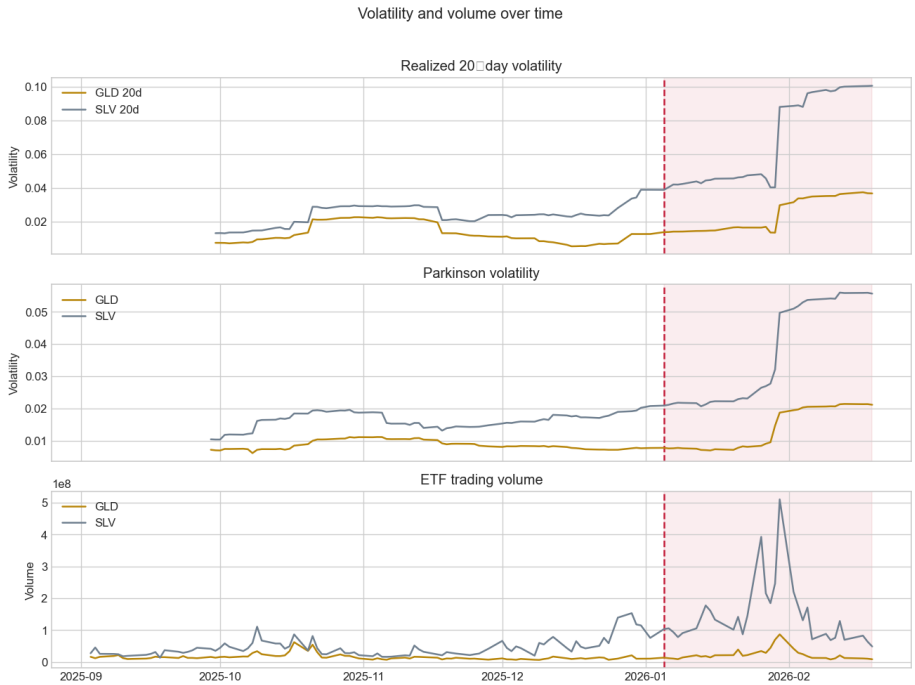
Cumulative returns (normalized to 100 at the first date): a clear view of relative performance and whether paths diverge after the event.



	mean	std	min	max
ret_gld	0.002946	0.019728	-0.108412	0.061647
ret_slv	0.005537	0.049015	-0.336037	0.086602
ret_xau_binance	0.003788	0.032585	-0.104457	0.064529
ret_xag_binance	0.000494	0.097086	-0.312222	0.145629

3.2 Volatility and liquidity over time

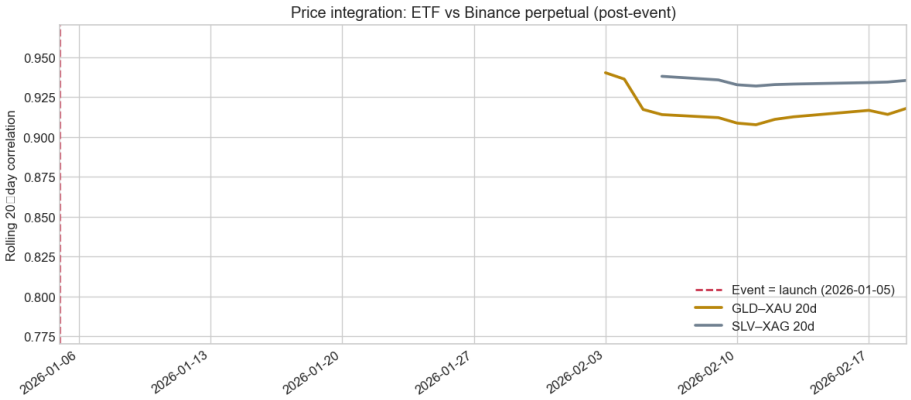
We look at 20-day realized volatility for GLD and SLV and ETF trading volume.



Why there is no pre-period for integration. The event is the *launch* of Binance XAUUSDT/XAGUSDT. Before that date there are no Binance prices or returns, so series like `corr_20_gld_xau` and `beta_60_gld_on_xau` (which need both ETF and Binance returns) are undefined before the launch. We therefore **cannot** estimate "integration before the event" in the same way we do for volatility or liquidity: there is no pre-period for Binance–ETF co-movement. We can only describe integration *after* the launch (level and evolution of rolling correlation) and, in the pre/post table, the "pre" mean for correlation is effectively missing or based on no data.

What this shows: The *rolling 20-day correlation* is the correlation between daily returns of the ETF (GLD or SLV) and the Binance perpetual (XAUUSDT or XAGUSDT) over the last 20 days. Values near 1 mean the two series move together. This metric exists only when both sides have data (post-event). The chart and the pre/post table in section 4 show post-event integration only; there is no comparable pre-event integration measure.

Single chart (gold and silver together): Same scale and dates so both lines are easy to compare. Y-axis is limited to the range where the data lies so the lines are not squeezed at the top. The vertical line is the **event date** — that is, the **launch date** of Binance XAUUSDT (2026-01-05). We use this single date as "the event" everywhere in the report. The SLV–XAG line starts later than GLD–XAU because XAGUSDT was listed a few days after XAUUSDT, so the 20-day rolling window has enough data only from a later date.



Minute-level correlation (last 7 days)

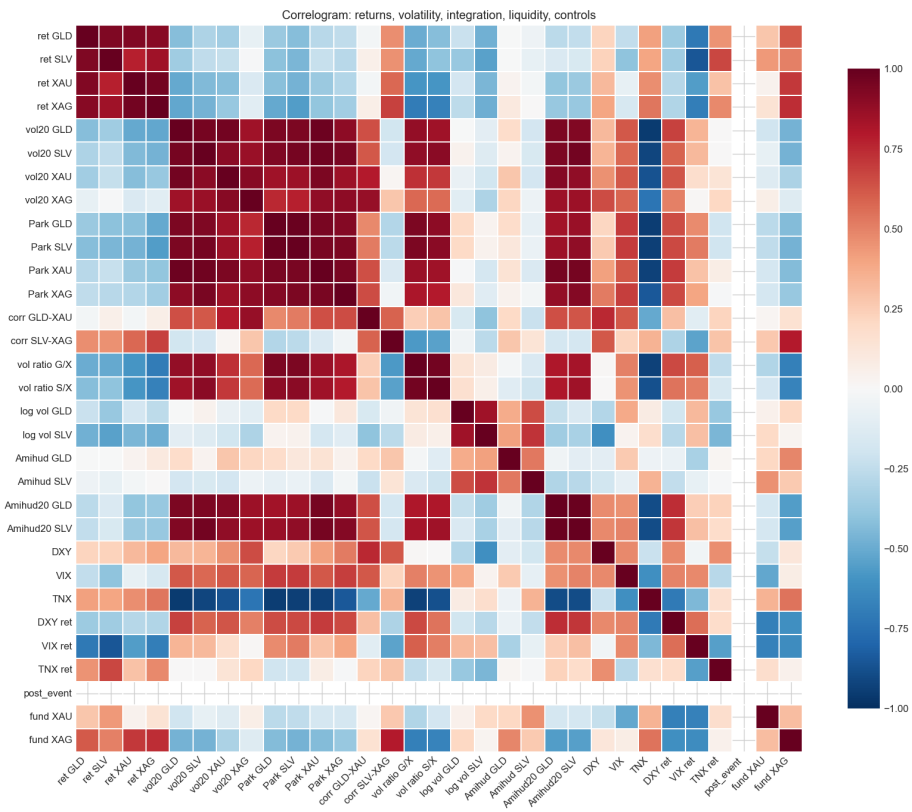
For the most recent 7 days we have *minute-level* data for both ETFs (Yahoo) and Binance. We merge on datetime and compute 1m returns to get a same-frequency measure of short-run co-movement.

```
Minute-level return correlation (last 7 days, overlapping 1m bars):
  GLD-XAU: 0.9913
  SLV-XAG: 0.9934
```

	mean	std	min	max
amihud_gld	7.483538e-10	6.085406e-10	3.474315e-12	2.885031e-09
amihud_20_gld	7.009107e-10	1.577653e-10	4.831240e-10	1.276925e-09
amihud_slv	4.780433e-10	3.276916e-10	1.153311e-12	2.008110e-09
amihud_20_slv	4.779806e-10	1.142224e-10	2.962718e-10	7.452803e-10

3.3 Correlogram

We compute the correlation matrix for a set of variables that are meaningful for the analysis: returns, volatility measures, integration (rolling correlation/beta), liquidity proxies, controls and the event dummy. Only numeric series with sufficient non-missing overlap are included.

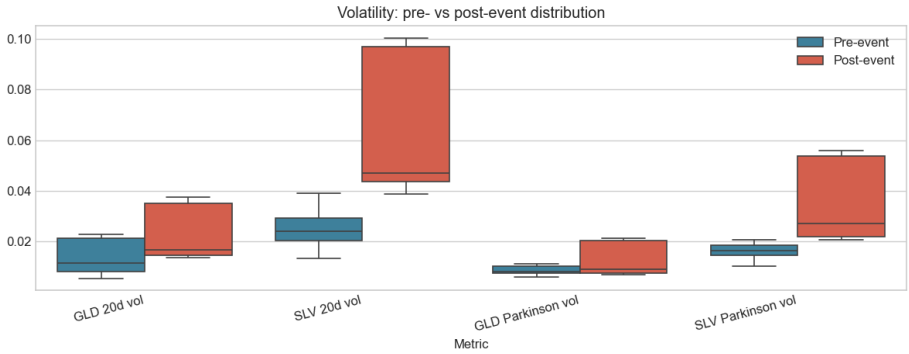


Comment. Variables with the largest absolute correlation with `post_event` are those that move most with the event. This helps spot which outcomes (e.g. volatility, correlation) are most linked to the launch.

4. Event-study style comparison

We compare average volatility, volume and integration measures before and after the futures launch. The event date is 2026-01-05 (Binance XAUUSDT TRADFI launch).

Pre vs post distributions for key metrics (boxplots by period):



	metric	period	mean	std	n
0	GLD 20d vol	pre	1.343866e-02	6.062648e-03	66
1	GLD 20d vol	post	2.380297e-02	1.011750e-02	32
2	SLV 20d vol	pre	2.411353e-02	6.133201e-03	66
3	SLV 20d vol	post	6.651057e-02	2.643642e-02	32
4	GLD Parkinson vol	pre	8.671084e-03	1.418413e-03	67
5	GLD Parkinson vol	post	1.357468e-02	6.393080e-03	32
6	SLV Parkinson vol	pre	1.625328e-02	2.567734e-03	67
7	SLV Parkinson vol	post	3.669129e-02	1.564470e-02	32
8	GLD volume	pre	1.512372e+07	9.048477e+06	85
9	GLD volume	post	2.289424e+07	1.756732e+07	32
10	SLV volume	pre	4.388922e+07	2.724932e+07	85
11	SLV volume	post	1.439370e+08	9.608540e+07	32
12	corr_20_gld_xau	pre	NaN	NaN	0
13	corr_20_gld_xau	post	9.173261e-01	1.028615e-02	12
14	corr_20_slv_xag	pre	NaN	NaN	0
15	corr_20_slv_xag	post	9.342025e-01	1.902024e-03	9

Comment. Compare the "pre" and "post" rows for each metric. If the post mean is higher than the pre mean for volatility or correlation, that points to an increase after the event. For volume, higher post means more trading. For liquidity (Amihud), lower post means better liquidity.

For a more formal comparison we can run simple t-tests.

	metric	t_stat	p_value
0	GLD 20d vol	5.347993	3.390943e-06
1	SLV 20d vol	8.955992	2.629522e-10
2	GLD Parkinson vol	4.288773	1.510642e-04
3	SLV Parkinson vol	7.342943	2.473673e-08
4	GLD volume	2.385871	2.221668e-02

	metric	t_stat	p_value
5	SLV volume	5.802936	1.743247e-06
6	corr_20_gld_xau	NaN	NaN
7	corr_20_slv_xag	NaN	NaN

Comment. A low p-value (e.g. below 0.05) means the difference between pre and post is statistically significant. Use this together with the pre/post means to say whether the change is meaningful.

5. Simple regression models

We estimate simple linear models to quantify the impact of the event and controls on ETF volatility and absolute returns.

Date	vol_realized_20_ret_gld	vol_realized_20_ret_slv	dxy_ret	vix_ret	tnx_ret	post
2025-09-30	0.007630	0.013341	-0.001431	0.009877	0.001689	0
2025-10-01	0.007610	0.013404	-0.000614	0.000614	-0.010177	0
2025-10-02	0.007558	0.013260	0.001432	0.020657	-0.004393	0
2025-10-03	0.007311	0.013733	-0.001329	0.001202	0.007555	0
2025-10-06	0.007869	0.013770	0.003983	-0.016960	0.010385	0

```
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0134      0.001     14.351      0.000       0.012       0.015
post_event     0.0104      0.002      6.307      0.000       0.007       0.014
dxy_ret        0.4462      0.248      1.801      0.075      -0.046       0.938
vix_ret        0.0051      0.010      0.518      0.606      -0.014       0.025
tnx_ret       -0.0515      0.097     -0.531      0.597      -0.244       0.141
==============================================================================
```

```
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0241      0.002     12.391      0.000       0.020       0.028
post_event     0.0424      0.003     12.436      0.000       0.036       0.049
dxy_ret        0.9825      0.515      1.909      0.059      -0.040       2.005
vix_ret        0.0028      0.020      0.135      0.893      -0.038       0.043
tnx_ret       -0.1760      0.201     -0.874      0.384      -0.576       0.224
==============================================================================
```

Comment. The `post_event` coefficient is the main result: it shows how much volatility changed after the launch, once we control for DXY, VIX and 10Y yield

changes. If it is positive and the p-value (P>|t|) is below 0.05, we conclude that volatility increased after the event (H1 supported for that ETF).

If needed, we can also model absolute returns as a simple proxy for volatility:

```
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0097      0.002      6.304      0.000       0.007       0.013
post_event     0.0120      0.003      4.048      0.000       0.006       0.018
dxy_ret        0.6152      0.417      1.474      0.143      -0.212       1.442
vix_ret        0.0082      0.018      0.459      0.647      -0.027       0.044
tnx_ret       -0.0755      0.164     -0.459      0.647      -0.401       0.250
==============================================================================
```

```
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0211      0.004      5.594      0.000       0.014       0.029
post_event     0.0354      0.007      4.900      0.000       0.021       0.050
dxy_ret        1.9819      1.022      1.940      0.055      -0.043       4.006
vix_ret        0.0543      0.044      1.241      0.217      -0.032       0.141
tnx_ret       -0.0995      0.402     -0.247      0.805      -0.897       0.698
==============================================================================
```

6. Interpretation and next steps

6.1 Key results summary

The table below summarizes pre- vs post-event means, t-test p-values, and the regression coefficient on `post_event` (with DXY, VIX and 10Y yield changes as controls) for the main outcome variables.

	Metric	Pre mean	Post mean	t-test p	post_event coef	regression p
0	GLD 20d vol	0.013439	2.380300e-02	0.0	0.010356	0.0
1	SLV 20d vol	0.024114	6.651100e-02	0.0	0.042416	0.0
2	GLD Parkinson vol	0.008671	1.357500e-02	0.0002	—	—
3	SLV Parkinson vol	0.016253	3.669100e-02	0.0	—	—
4	GLD volume	15123724.705882	2.289424e+07	0.0222	—	—
5	SLV volume	43889215.294118	1.439370e+08	0.0	—	—
6	Corr GLD–XAU	—	9.173260e-01	—	—	—
7	Corr SLV–XAG	—	9.342020e-01	—	—	—

Comment. This table brings everything together. Use "Pre mean" vs "Post mean" to see the direction of change; "t-test p" to see if the change is significant; and "post_event coef" with "regression p" (for GLD and SLV volatility) to

see the effect of the event after controlling for macro factors. For volatility, a positive and significant post_event coef supports H1; for volume, higher post mean supports H2. Correlation columns (Corr GLD–XAU, Corr SLV–XAG) have no meaningful pre mean (Binance did not exist), so we use them only as descriptive post-event integration.

6.2 Empirical results

```
**Volatility (H1):** After controlling for DXY, VIX and 10Y yield changes, the post-event co

For SLV, the post_event coefficient is 0.04242 (p = 0.000): volatility increased and is sign

**Integration (descriptive only):** Average rolling correlation GLD-
XAU is 0.917 post-event; SLV-XAG is 0.934 post-event. There is no pre-event correlation (Bin

**Liquidity (H2):** Average GLD volume is 22894241 post-event vs 15123725 pre-event; SLV vol
```

6.3 Conclusions on hypotheses

Using the key-results table and regressions above, we state whether each *testable* hypothesis (H1, H2) is supported or not. Integration is not tested as a hypothesis (no pre-period); we only describe post-event correlation.

```
H1 (volatility increase after launch):
  GLD: supported (positive and significant post_event means higher volatility).
  SLV: supported.

Integration (descriptive, post-event only; no pre/post test):
  Rolling correlation GLD-XAU and SLV-XAG are reported in section 3.2 and the key-results ta
  We do not test a formal integration hypothesis because Binance data do not exist before th

H2 (improved liquidity: higher volume, lower Amihud):
  GLD volume: higher post (higher post supports H2).
  SLV volume: higher post.
  Check Amihud in the pre/post table; lower post = better liquidity.
```

6.4 Limitations and extensions

- **No pre-period for integration.** Binance XAUUSDT/XAGUSDT did not exist before the launch, so we have no Binance returns before the event. Rolling correlation (GLD–XAU, SLV–XAG) and beta can only be computed from the launch date onward. We cannot compare "integration before vs after" in the same way as for volatility or liquidity; we can only describe the level and evolution of integration post-launch.
- **Short post-event window.** We have limited days after the Binance launch. Results can change when more data are available.

- **Robustness.** Try different event dates, volatility windows (e.g. 10d or 60d), or placebo tests (fake event date) to see if the effects hold.
- **Beta.** The key-results table focuses on volatility, volume and correlation. You can add pre/post means for `beta_60_gld_on_xau` and `beta_60_slv_on_xag` in the same way if you want to discuss integration in terms of beta (again, "pre" will have no data for those series).

6.5 Conclusion

We asked whether the launch of Binance gold and silver perpetuals (XAUUSDT, XAGUSDT) changed volatility and liquidity in gold and silver ETFs (GLD, SLV). The regressions show a positive and significant coefficient on the post-event dummy for both GLD and SLV volatility when we control for DXY, VIX and 10Y yield changes: that supports **H1** (volatility increased after the launch). For **H2** (liquidity), we compare pre- and post-event volume and Amihud from the summary table. We do not test a formal hypothesis on integration (no pre-event Binance data); we only report post-event rolling correlation as descriptive evidence. In short: the evidence points to higher ETF volatility after the Binance TRADFI launch; liquidity and integration are discussed in the tables and section 6.3. Because the post-event window is still short, updating the analysis with more data is recommended.